

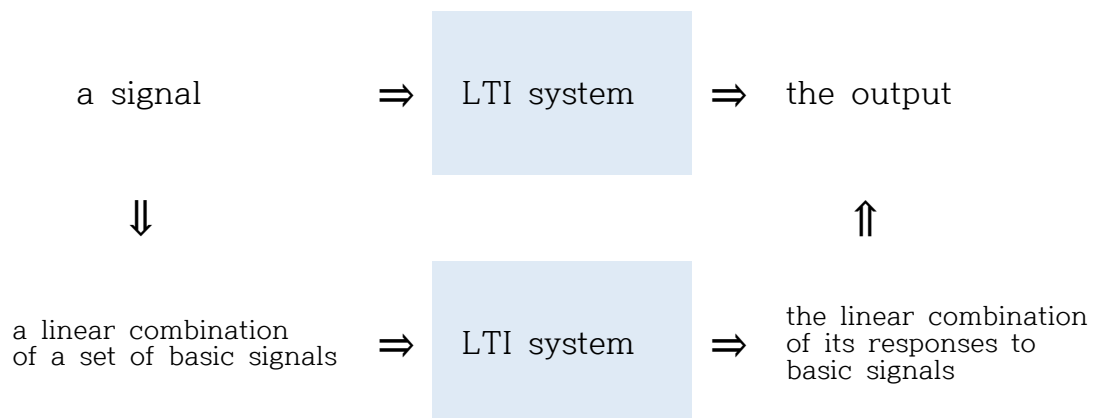
Chap. 2. Linear Time Invariant Systems

- Linear Time-Invariant (LTI) System
- Impulse Response
- Sifting Property
- Convolution Sum / Convolution Integral
- Linear Constant-Coefficient Differential Equation (LCCDE)
 Linear Constant-Coefficient Difference Equation (LCCDE)
- Infinite Impulse Response (IIR) System
 Finite Impulse Response (FIR) System
- Recursive System / Nonrecursive System
 Feedback System / Feedforward System
- Block Diagram
- State-Variable Description

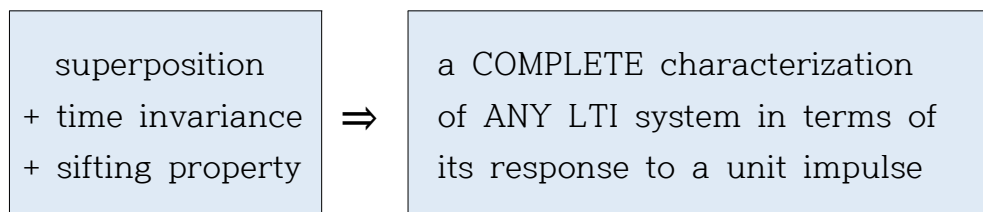
2.0 Introduction

● Linear Time-Invariant Systems

- Many physical processes can be modeled as LTI systems.
- a set of powerful tools that form the core of signal and system analysis
- Superposition property (ex: circuit theory)



- General signals can be represented as linear combinations of delayed impulses. (sifting property)
- Convolution



- linear constant-coefficient differential equations
- linear constant-coefficient difference equations

2.1 Discrete-time LTI systems

2.1.1 Discrete-time signals in terms of impulses

- Weighted time-shift impulses

$$x[-1]\delta[n+1] = \begin{cases} x[-1], & n = -1 \\ 0, & n \neq -1 \end{cases}$$

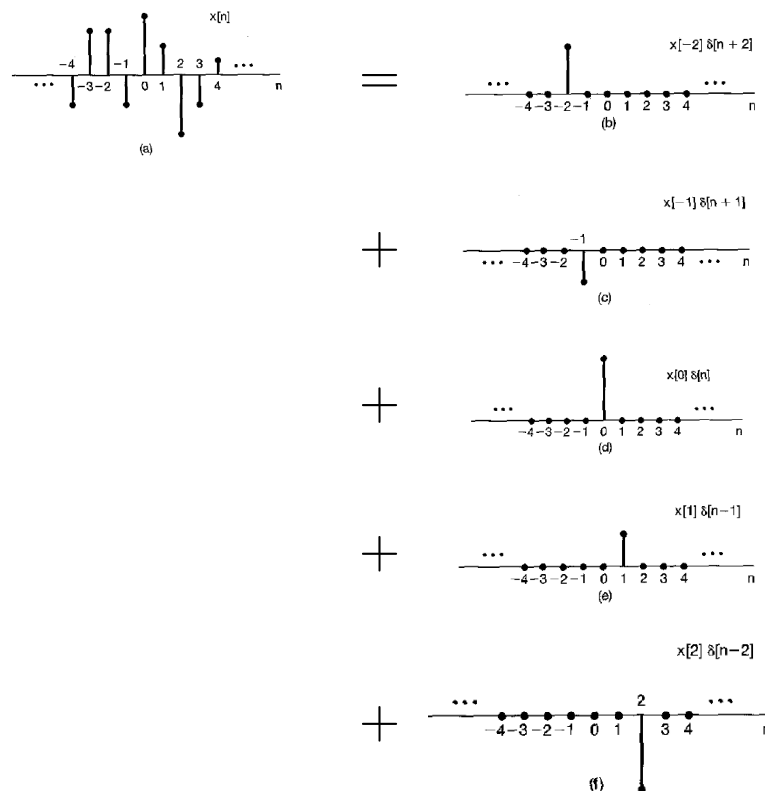
$$x[0]\delta[n] = \begin{cases} x[0], & n = 0 \\ 0, & n \neq 0 \end{cases}$$

$$x[1]\delta[n-1] = \begin{cases} x[1], & n = 1 \\ 0, & n \neq 1 \end{cases}$$

- More generally, $x[n]$ can be written:

$$\begin{aligned} x[n] &= \cdots + x[-2]\delta[n+2] + x[-1]\delta[n+1] \\ &\quad + x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + \cdots \\ &= \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \end{aligned} \quad (2.2)$$

- Decomposition of a discrete-time signal into a weighted sum of shifted impulses (Fig. 2.1)



- Eq(2.2): SIFTING property (NOT shifting property)
An arbitrary signal can be represented as a linear combination of shifted unit impulses $\delta[n-k]$, where the weights are $x[k]$.

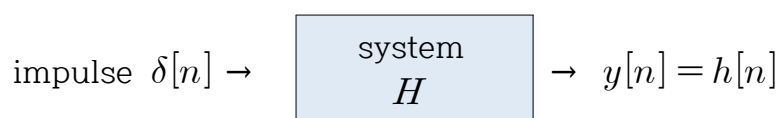
$$\ast \text{ [Importance of Eq.(2.2) } x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] \text{]}$$

- Any discrete-time signal can be represented by using Eq. (2.2). [i.e., the weighted sum of time-shifted impulses]
- The impulse is the basic element of signals.
- A signal can be represented
a summation [$\ast$ linear (additivity)]
of the weighted [$\ast$ linear (homogeneity)]
time-shifted impulses. [$\ast$ time-invariant]
- linear combination (i.e., a kind of series)

2.1.2 The convolution sum representation of LTI systems

- IMPULSE RESPONSE

the output of a system $h[n]$, when the impulse $\delta[n]$ is applied to the system as the input



- the **importance** of the impulse response

If we know the impulse response of an LTI system,
we can get the response (i.e., output) of the system
for any input by using Eq. (2.2) and Eq. (2.6).

- (Given) the impulse response of an LTI system : $h[n]$

$$\delta[n] \Rightarrow h[n] \quad (\text{known})$$

$$\begin{aligned} \dots &\Rightarrow \dots \\ \delta[n] &\Rightarrow h[n] && (\text{known}) \\ \delta[n-1] &\Rightarrow h[n-1] && (\text{time invariant}) \\ \delta[n-2] &\Rightarrow h[n-2] && (\text{time invariant}) \\ \dots &\Rightarrow \dots \end{aligned}$$

$$\begin{aligned} \dots &\Rightarrow \dots \\ a_0 \delta[n] &\Rightarrow a_0 h[n] && (\text{linear}) \\ a_1 \delta[n-1] &\Rightarrow a_1 h[n-1] && (\text{linear}) \\ a_2 \delta[n-2] &\Rightarrow a_2 h[n-2] && (\text{linear}) \\ \dots &\Rightarrow \dots \end{aligned}$$

$$\begin{aligned} \dots + \delta[n] + \delta[n-1] + \delta[n-2] + \dots \\ \Rightarrow \dots + h[n] + h[n-1] + h[n-2] + \dots \quad (\text{linear}) \end{aligned}$$

$$\begin{aligned} \dots + a_0 \delta[n] + a_1 \delta[n-1] + a_2 \delta[n-2] + \dots \\ \Rightarrow \dots + a_0 h[n] + a_1 h[n-1] + a_2 h[n-2] + \dots \quad (\text{linear}) \end{aligned}$$

$$\begin{aligned} x[n] &= \dots + a_0 \delta[n] + a_1 \delta[n-1] + a_2 \delta[n-2] + \dots \\ &= \dots + x[0] \delta[n] + x[1] \delta[n-1] + x[2] \delta[n-2] + \dots \\ &= \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \quad (\text{linear combination of shifted impulses}) \end{aligned}$$

$$\begin{aligned} y[n] &= \dots + a_0 h[n] + a_1 h[n-1] + a_2 h[n-2] + \dots \\ &= \dots + x[0] h[n] + x[1] h[n-1] + x[2] h[n-2] + \dots \\ &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] \quad (\text{convolution sum}) \end{aligned}$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \quad (\text{sifting property}) \quad (2.2)$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] \quad (\text{convolution sum}) \quad (2.6)$$

- the symbol of convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] \equiv x[n] * h[n] \quad (2.7)$$

[Example 2.1]

output: $y[n] = x[0]h[n-0] + x[1]h[n-1] = 0.5h[n] + 2h[n-1]$

$$n = -2 \quad \dots\dots\dots$$

$$n = -1 \quad y[-1] = x[0]h[-1-0] + x[1]h[-1-1] = 0.5h[-1] + 2h[-1-1]$$

$$n = 0 \quad y[0] = x[0]h[0-0] + x[1]h[0-1] = 0.5h[0] + 2h[0-1]$$

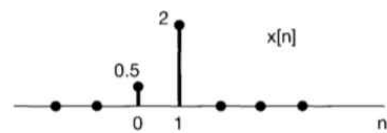
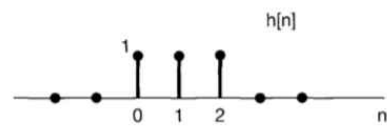
$$n = 1 \quad y[1] = x[0]h[1-0] + x[1]h[1-1] = 0.5h[1] + 2h[1-1]$$

$$n = 2 \quad y[2] = x[0]h[2-0] + x[1]h[2-1] = 0.5h[2] + 2h[2-1]$$

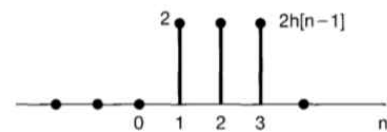
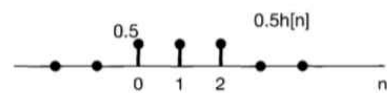
$$n = 3 \quad y[3] = x[0]h[3-0] + x[1]h[3-1] = 0.5h[3] + 2h[3-1]$$

$$n = 4 \quad y[4] = x[0]h[4-0] + x[1]h[4-1] = 0.5h[4] + 2h[4-1]$$

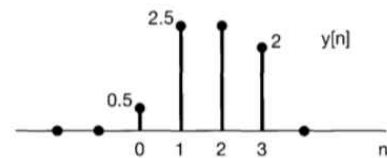
$$n = 5 \quad \dots\dots\dots$$



(a)

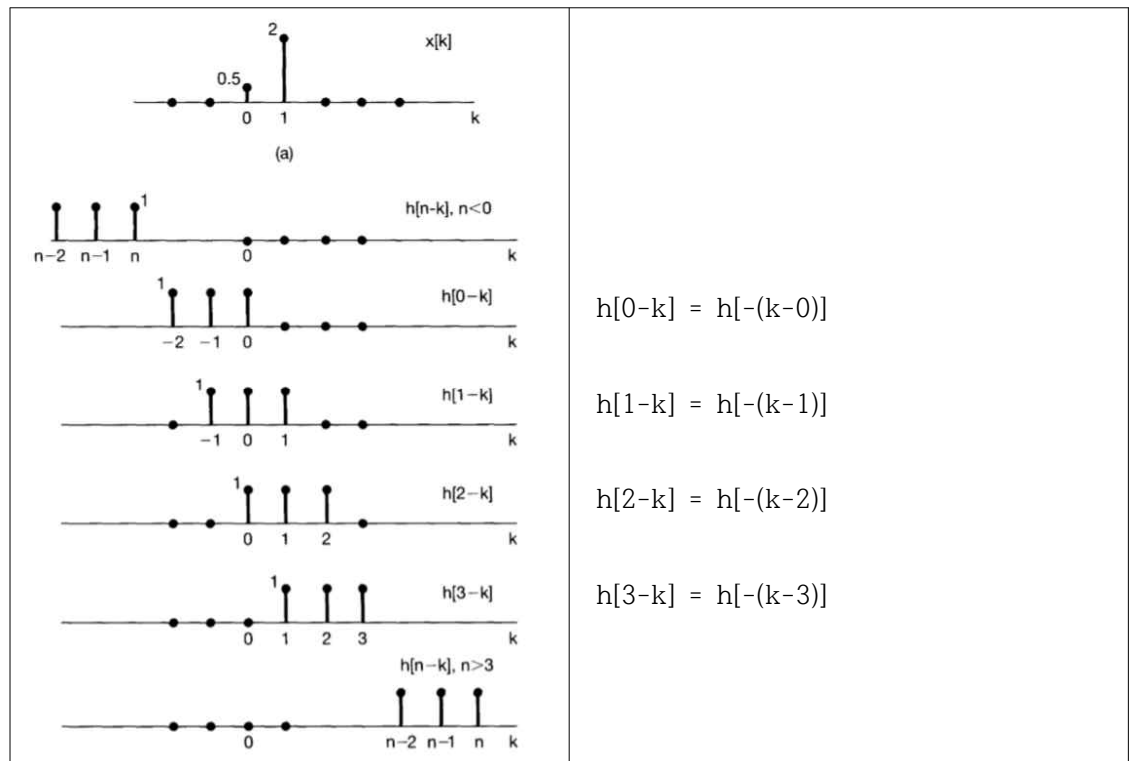


(b)



(c)

[Example 2.2] Consider Example 2.1 again. (other method)



※ output in Example 2.1

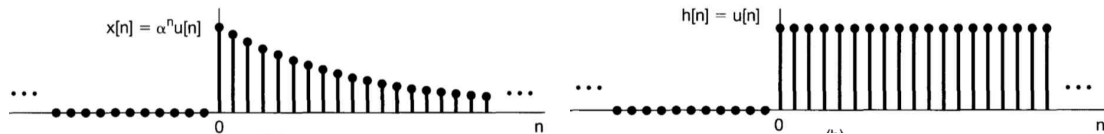
.....
$y[0] = \sum_{k=-\infty}^{\infty} x[k]h[0-k] = 0.5.$
$y[1] = \sum_{k=-\infty}^{\infty} x[k]h[1-k] = 0.5 + 2.0 = 2.5.$
$y[2] = \sum_{k=-\infty}^{\infty} x[k]h[2-k] = 0.5 + 2.0 = 2.5,$
$y[3] = \sum_{k=-\infty}^{\infty} x[k]h[3-k] = 2.0.$
.....

[Example 2.3]

$$x[n] = \alpha^n u[n] \quad (0 < \alpha < 1), \quad h[n] = u[n]$$

$$x[k]h[n-k] = \begin{cases} \alpha^k, & 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}$$

$$y[n] = \sum_{k=0}^n x[k]h[n-k] = \sum_{k=0}^n \alpha^k = \left(\frac{1-\alpha^{n+1}}{1-\alpha} \right) u[n]$$



$x[k]$	
$h[-k]$	
$h[-(k+1)]$ $= h[-1-k]$	
$h[-(k-1)]$ $= h[1-k]$	
$h[-(k-n)]$ $= h[n-k] \quad n > 0$	
$h[-(k-n)]$ $= h[n-k] \quad n < 0$	
$y[n] = \left(\frac{1-\alpha^{n+1}}{1-\alpha} \right) u[n]$	

[Example]

$$x[n] = \begin{cases} 2, & n=0 \\ 3, & n=1 \\ -2, & n=2 \\ 0, & \text{otherwise} \end{cases} \quad h[n] = \begin{cases} 1, & n=-1 \\ 2, & n=0 \\ 2, & n=1 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{convolution sum} \quad y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

.....

$$\begin{aligned} y[-2] &= \dots + x[-1]h[-1] + x[0]h[-2] + x[1]h[-3] + x[2]h[-4] + x[3]h[-5] + \dots \\ &= \dots + 0 \times 1 + 2 \times 0 + 3 \times 0 + (-2) \times 0 + 0 \times 0 + \dots = 0 \end{aligned}$$

$$\begin{aligned} y[-1] &= \dots + x[-1]h[0] + x[0]h[-1] + x[1]h[-2] + x[2]h[-3] + x[3]h[-4] + \dots \\ &= \dots + 0 \times 2 + 2 \times 1 + 3 \times 0 + (-2) \times 0 + 0 \times 0 + \dots = 2 \end{aligned}$$

$$\begin{aligned} y[0] &= \dots + x[-1]h[1] + x[0]h[0] + x[1]h[-1] + x[2]h[-2] + x[3]h[-3] + \dots \\ &= \dots + 0 \times 2 + 2 \times 2 + 3 \times 1 + (-2) \times 0 + 0 \times 0 + \dots = 7 \end{aligned}$$

$$\begin{aligned} y[1] &= \dots + x[-1]h[2] + x[0]h[1] + x[1]h[0] + x[2]h[-1] + x[3]h[-2] + \dots \\ &= \dots + 0 \times 0 + 2 \times 2 + 3 \times 2 + (-2) \times 1 + 0 \times 0 + \dots = 8 \end{aligned}$$

$$\begin{aligned} y[2] &= \dots + x[-1]h[3] + x[0]h[2] + x[1]h[1] + x[2]h[0] + x[3]h[-1] + \dots \\ &= \dots + 0 \times 0 + 2 \times 0 + 3 \times 2 + (-2) \times 2 + 0 \times 1 + \dots = 2 \end{aligned}$$

$$\begin{aligned} y[3] &= \dots + x[-1]h[4] + x[0]h[3] + x[1]h[2] + x[2]h[1] + x[3]h[0] + \dots \\ &= \dots + 0 \times 0 + 2 \times 0 + 3 \times 0 + (-2) \times 2 + 0 \times 2 + \dots = -4 \end{aligned}$$

$$\begin{aligned} y[4] &= \dots + x[-1]h[5] + x[0]h[4] + x[1]h[3] + x[2]h[2] + x[3]h[1] + \dots \\ &= \dots + 0 \times 0 + 2 \times 0 + 3 \times 0 + (-2) \times 0 + 0 \times 2 + \dots = 0 \end{aligned}$$

.....

$$\therefore y[n] = \{ \dots, 0, 0, 2, 7, 8, 2, -4, 0, 0, \dots \}$$

$\uparrow$
 $n=0$

(Example) Graphical Method

$$\begin{aligned}
x[n] &= \dots, 0, 0, 0, 2, 3, -2, 0, 0, \dots \\
h[n] &= \dots, 0, 0, 1, 2, 2, 0, 0, 0, \dots \\
h[-n] &= \dots, 0, 0, 2, 2, 1, 0, 0, 0, \dots \\
h[1-n] &= h[-(n-1)] = \dots, 0, 0, 0, 2, 2, 1, 0, 0, 0, \dots
\end{aligned}$$

$$\begin{aligned}
y[-2] &= \sum_{k=-\infty}^{\infty} x[k]h[-2-k] = \sum_{k=-\infty}^{\infty} x[k]h[-(k+2)] \\
\begin{array}{rcl}
x[k] & = & \dots, 0, 0, 0, 0, 2, 3, -2, 0, 0, 0, \dots \\
h[-(k+2)] & = & \dots, 0, 2, 2, 1, 0, 0, \quad 0, 0, 0, 0, \dots
\end{array} \\
\hline
& \dots, 0, 0, 0, 0, 0, 0, \quad 0, 0, 0, 0, \dots \\
\therefore y[-2] &= 0
\end{aligned}$$

$$\begin{aligned}
y[-1] &= \sum_{k=-\infty}^{\infty} x[k]h[-1-k] = \sum_{k=-\infty}^{\infty} x[k]h[-(k+1)] \\
\begin{array}{rcl}
x[k] & = & \dots, 0, 0, 0, 0, 2, 3, -2, 0, 0, 0, \dots \\
h[-(k+1)] & = & \dots, 0, 0, 2, 2, 1, 0, \quad 0, 0, 0, 0, \dots
\end{array} \\
\hline
& \dots, 0, 0, 0, 0, 2, 0, \quad 0, 0, 0, 0, \dots \\
\therefore y[-1] &= 2
\end{aligned}$$

$$\begin{aligned}
y[0] &= \sum_{k=-\infty}^{\infty} x[k]h[0-k] = \sum_{k=-\infty}^{\infty} x[k]h[-(k+0)] \\
\begin{array}{rcl}
x[k] & = & \dots, 0, 0, 0, 0, 2, 3, -2, 0, 0, 0, \dots \\
h[-k] & = & \dots, 0, 0, 0, 2, 2, 1, \quad 0, 0, 0, 0, \dots
\end{array} \\
\hline
& \dots, 0, 0, 0, 0, 4, 3, \quad 0, 0, 0, 0, \dots \\
\therefore y[0] &= 4 + 3 = 7
\end{aligned}$$

$$\begin{aligned}
y[1] &= \sum_{k=-\infty}^{\infty} x[k]h[1-k] = \sum_{k=-\infty}^{\infty} x[k]h[-(k-1)] \\
\begin{array}{rcl}
x[k] & = & \dots, 0, 0, 0, 0, 2, 3, -2, 0, 0, 0, \dots \\
h[-(k-1)] & = & \dots, 0, 0, 0, 0, 2, 2, \quad 1, 0, 0, 0, \dots
\end{array} \\
\hline
& \dots, 0, 0, 0, 0, 4, 6, -2, 0, 0, 0, \dots \\
\therefore y[1] &= 4 + 6 - 2 = 8
\end{aligned}$$

$$\begin{aligned}
y[2] &= \sum_{k=-\infty}^{\infty} x[k]h[2-k] = \sum_{k=-\infty}^{\infty} x[k]h[-(k-2)] \\
x[k] &= \cdots, 0, 0, 0, 0, 2, 3, -2, 0, 0, 0, \cdots \\
h[-(k-2)] &= \cdots, 0, 0, 0, 0, 0, 2, \quad 2, 1, 0, 0, \cdots \\
\hline
&\quad \cdots, 0, 0, 0, 0, 0, 6, -4, 0, 0, 0, \cdots \\
&\therefore y[2] = 6 - 4 = 2
\end{aligned}$$

$$\begin{aligned}
y[3] &= \sum_{k=-\infty}^{\infty} x[k]h[3-k] = \sum_{k=-\infty}^{\infty} x[k]h[-(k-3)] \\
x[k] &= \cdots, 0, 0, 0, 0, 2, 3, -2, 0, 0, 0, \cdots \\
h[-(k-3)] &= \cdots, 0, 0, 0, 0, 0, 0, \quad 2, 2, 1, 0, \cdots \\
\hline
&\quad \cdots, 0, 0, 0, 0, 0, 0, -4, 0, 0, 0, \cdots \\
&\therefore y[3] = -4
\end{aligned}$$

$$\begin{aligned}
y[4] &= \sum_{k=-\infty}^{\infty} x[k]h[4-k] = \sum_{k=-\infty}^{\infty} x[k]h[-(k-4)] \\
x[k] &= \cdots, 0, 0, 0, 0, 2, 3, -2, 0, 0, 0, \cdots \\
h[-(k-4)] &= \cdots, 0, 0, 0, 0, 0, 0, \quad 0, 2, 2, 1, \cdots \\
\hline
&\quad \cdots, 0, 0, 0, 0, 0, 0, \quad 0, 0, 0, 0, \cdots \\
&\therefore y[4] = 0
\end{aligned}$$

$$\begin{aligned}
\therefore y[n] &= \{ \cdots, 0, 0, 2, 7, 8, 2, -4, 0, 0, \cdots \} \\
&\quad \uparrow \\
&\quad n = 0
\end{aligned}$$

§2.2 Continuous-time LTI Systems: The Convolution Integral

- The **convolution sum**, the output of the discrete-time LTI system, is derived by expressing the input signal $x[n]$ as a weighted sum of time-shifted impulses as shown by

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

- The **convolution integral**, the output of the continuous-time LTI system, is derived by expressing the input signal $x(t)$ as a weighted superposition of time-shifted impulses as shown by

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

.Impulse Response of the LTI continuous-time system

$$\text{impulse } \delta(t) \rightarrow \boxed{\begin{array}{c} \text{system} \\ H \end{array}} \rightarrow y(t) = h(t)$$

.output $y(t)$ for an input $x(t)$

$$x(t) \rightarrow \boxed{\begin{array}{c} \text{system} \\ H \end{array}} \rightarrow y(t) = H[x(t)]$$

$$\begin{aligned} y(t) &= H \left[\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau \right] \\ &= \int_{-\infty}^{\infty} x(\tau) H[\delta(t - \tau)] d\tau \quad (\text{linearity}) \\ &= \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad (\text{time invariant}) \\ &\equiv x(t) * h(t) \end{aligned}$$

	discrete-time	continuous-time
sifting property	$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$	$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$
convolution sum/integral	$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$	$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$
	Eq.(2.2) / Eq.(2.6)	Eq.(2.27) / Eq.(2.33)

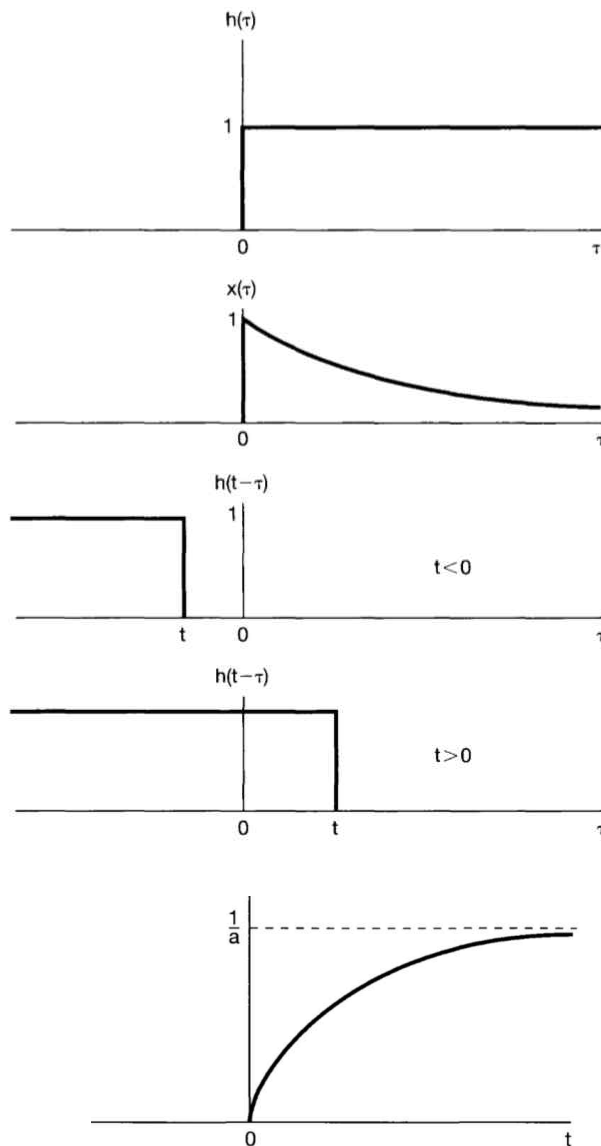
[Example 2.6]

$$x(t) = e^{-at} u(t), \quad a > 0 \quad h(t) = u(t)$$

$$x(\tau)h(t-\tau) = \begin{cases} e^{-a\tau}, & 0 < \tau < t \\ 0, & \text{otherwise} \end{cases}$$

$$y(t) = \int_0^t e^{-a\tau} d\tau = \frac{1}{a}(1 - e^{-at}), \quad t > 0$$

$$\therefore y(t) = \frac{1}{a}(1 - e^{-at}) u(t)$$



※ Why we need to use “Convolution” to get the output of a system?

[example] a circuit with resistors

§2.3 Properties of LTI Systems

- Convolution sum/integral

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n] \quad (2.39)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = x(t) * h(t) \quad (2.40)$$

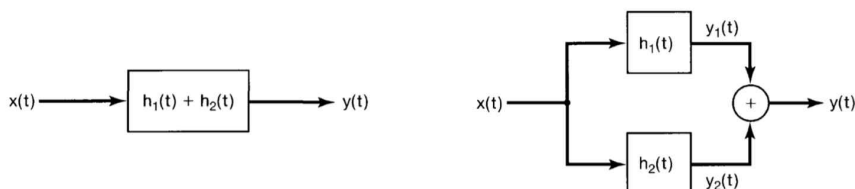
- Eq.(2.39) and (2.40) tell us that the characteristics of an LTI system are COMPLETELY determined by its impulse response.
- The impulse response of a nonlinear system does not completely characterize the behavior of the system.

§2.3.1 The commutative property

$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] = x[n] * h[n] \\ &= \sum_{k=-\infty}^{\infty} h[k] x[n-k] = h[n] * x[n] \\ y(t) &= \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = x(t) * h(t) \\ &= \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau = h(t) * x(t) \end{aligned}$$

§2.3.2 The distributive property

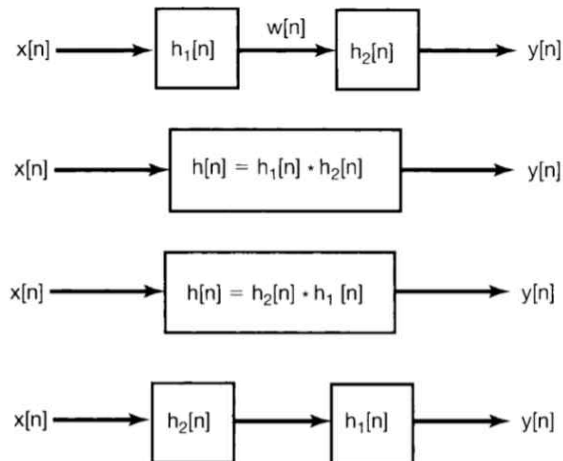
$$\begin{aligned} x[n] * \{h_1[n] + h_2[n]\} &= x[n] * h_1[n] + x[n] * h_2[n] \\ x(t) * \{h_1(t) + h_2(t)\} &= x(t) * h_1(t) + x(t) * h_2(t) \end{aligned}$$



§2.3.3 The associative property

$$x[n] * \{h_1[n] * h_2[n]\} = \{x[n] * h_1[n]\} * h_2[n]$$

$$x(t) * \{h_1(t) * h_2(t)\} = \{x(t) * h_1(t)\} * h_2(t)$$



§2.3.4 LTI systems with and without Memory

- Memoryless system

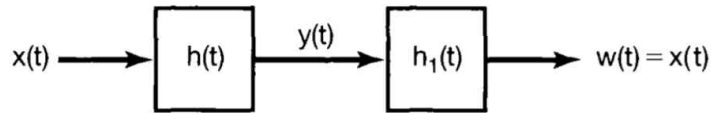
The output at any time depends ONLY on the value of the input at that same time.

$h[n] = 0 \quad \text{for} \quad n \neq 0$	$h(t) = 0 \quad \text{for} \quad t \neq 0$
$h[n] = K \delta[n]$	$h(t) = K \delta(t)$
$y[n] = K x[n]$	$y(t) = K x(t)$

- ※ In this case, the convolution sum/integral is also satisfied.
- ※ convolution property is for the memory system
(Memoryless systems are special cases of memory systems)

§2.3.5 Invertibility of LTI systems

- A system is invertible if the input to the system can be recovered from the output. [(ex) transmitter/receiver]



$$\begin{aligned} \text{If } w(t) &= y(t) * h_1(t) = [x(t) * h(t)] * h_1(t) \\ &= x(t) * [h(t) * h_1(t)] = x(t), \end{aligned}$$

then $h(t) * h_1(t) = \delta(t)$.

$h_1(t)$ is the inverse system of $h(t)$.

- ※ The inverse system of a system can not be existed.
zeros/poles of a system $\Leftrightarrow$ poles/zeros of the inverse system

- ※ Convolution :

Find output $[y(t)]$ from input $[x(t)]$ and system $[h(t)]$

- ※ Deconvolution :

Find input $[x(t)]$ from output $[y(t)]$ and system $[h(t)]$

(ex) estimate of the seismic center [input]
from the measured seismic waves [output]
and properties of the earth [system]

§2.3.6 Causality for LTI systems

- Memoryless systems
(present input) $\Rightarrow$ (present output)
- Ideal memory systems
(past + present + future input) $\Rightarrow$ (present output)
※ Reality?
- Causal systems
(past + present input) $\Rightarrow$ (present output)
(Example) Memoryless systems are causal.

- The output of a causal system depends only on the present and past values of the input to the system. $y[n]$ must not depend on $x[k]$ for $k > n$ (i.e., inputs in the future).
- From Eq.(2.39), $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$, we see that for this to be true, all of the coefficients $h[n-k]$ that multiply values of $x[k]$ for $k > n$ must be zero.

$$\boxed{\therefore h[n] = 0 \text{ for } n < 0} \quad (2.77)$$

- For a causal discrete-time LTI system,

$$y[n] = \sum_{k=-\infty}^n x[k] h[n-k] = \sum_{k=0}^{\infty} h[k] x[n-k] \quad (2.79)$$

Similarly, for a causal continuous-time LTI system,

$$h(t) = 0 \quad \text{for } t < 0 \quad (2.80)$$

$$y(t) = \int_{-\infty}^t x(\tau) h(t-\tau) d\tau = \int_0^{\infty} h(\tau) x(t-\tau) d\tau \quad (2.81)$$

§2.3.7 Stability for LTI systems

BIBO (Bounded Input Bounded Output) stable

$y[n]$ or $y(t)$: bounded (i.e., stable systems), if

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty \quad (\text{absolutely summable}) \quad (2.86)$$

$$\text{or} \quad \int_{-\infty}^{+\infty} |h(\tau)| d\tau < \infty \quad (\text{absolutely integrable}) \quad (2.87)$$

§2.3.8 The response of an LTI system

- the unit step response

$s[n] = u[n] * h[n] = \sum_{k=-\infty}^n h[k]$ $\therefore h[n] = s[n] - s[n-1]$	$s(t) = u(t) * h(t) = \int_{-\infty}^t h(\tau) d\tau$ $\therefore h(t) = \frac{ds(t)}{dt}$
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- Sinusoidal steady-state response

discrete-time LTI systems	continuous-time LTI system
$x[n] = e^{j\omega n} \rightarrow \boxed{h[n]} \rightarrow y[n]$	$x(t) = e^{j\omega t} \rightarrow \boxed{h(t)} \rightarrow y(t)$
$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$ $= \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)}$ $= e^{j\omega n} \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}$ $\equiv H(e^{j\omega}) e^{j\omega n}$	$y(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$ $= \int_{-\infty}^{\infty} h(\tau) e^{j\omega(t-\tau)} d\tau$ $= e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega \tau} d\tau$ $\equiv H(j\omega) e^{j\omega t}$
$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}$ NOT a function of time n, BUT only a function of frequency ω $\therefore$ frequency response of the discrete-time LTI system	$H(j\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega \tau} d\tau$ NOT a function of time t, BUT only a function of frequency ω $\therefore$ frequency response of the continuous-time LTI system

§2.4 Causal LTI Systems described by Differential and Difference Eq.

§2.4.1 Linear constant-coefficient differential equations (LCCDE)

General N-th order LCCDE

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k} \quad (2.109)$$

(Example) First-order LCCDE

$$\frac{dy(t)}{dt} + 2y(t) = x(t) \quad (\text{ex. RC circuit})$$

(Example) Second-order LCCDE (many control systems)

$$\frac{1}{C} y(t) + R \frac{dy(t)}{dt} + L \frac{d^2 y(t)}{dt^2} = \frac{dx(t)}{dt} \quad (\text{ex. RLC circuit})$$

2.4.2 Linear constant-coefficient difference equations (LCCDE)

General N-th order LCCDE

[the discrete-time counterpart of Eq. (2.109)]

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k] \quad (2.113)$$

(example)

$$y[n] + y[n-1] + \frac{1}{4}y[n-2] = x[n] + 2x[n-1]$$

- Recursive equation : Eq. (2.113) can be rearranged

$$y[n] = \frac{1}{a_0} \left\{ \sum_{k=0}^M b_k x[n-k] - \sum_{k=1}^N a_k y[n-k] \right\} \quad (2.115)$$

※ an Infinite Impulse Response (IIR) system

- Nonrecursive equation

$$y[n] = \sum_{k=0}^M \left(\frac{b_k}{a_0} \right) x[n-k] \quad (2.116)$$

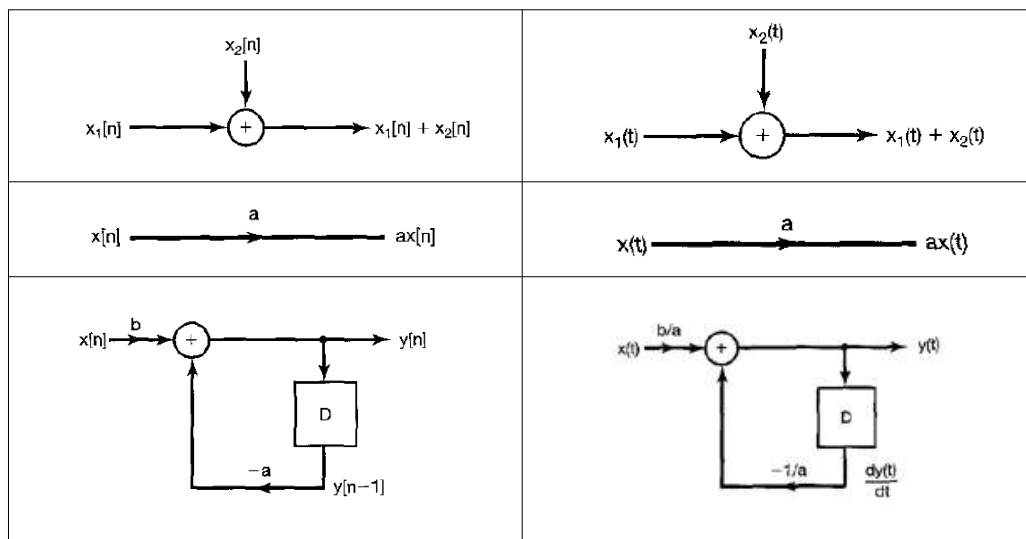
※ an Finite Impulse Response (FIR) system

(Example)

$$\sum_{k=0}^N a_k \frac{d^k}{dt^k} y(t) = 0 \rightarrow y(t) \quad (* \text{ Engineering Math})$$

$$\sum_{k=0}^N a_k y[n-k] = 0 \rightarrow y[n] \quad (* \text{ recurrence formula in Calculus})$$

§2.4.3 Block Diagram Representations



§2.5 Singularity Functions ... skip

§2.6 Summary

- system representation methods in the time domain
 1. impulse response
 2. linear constant-coefficient differential equation
 3. block diagram
 4. state-variable description
- sifting property
 - weighted sums [or integrals] of shifted unit impulses
- convolution sum [or integral]
- The sifting and convolution properties are extremely important, as they allow us to compute the response of an LTI system to an arbitrary input in terms of the system's response to a unit impulse.